

海洋熱浪.

Marine Heatwaves

葉庭禎

熱浪事件

'Unprecedented' alerts in France as blistering heat grips Europe

13 hours ago

2025/07/01 歐洲 西班牙

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Kathryn Armstrong
BBC News



Watch: 'A little bit melting' - Intense heat across Europe

<https://www.bbc.com/news/articles/c5y7781e915o>

預防高溫熱傷害 北市府啟動今年首次熱浪預警機制

發布機關：臺北市政府環境保護局氣候變遷管理科

承辦單位：氣候變遷管理科 114/06/30 臺灣 臺北

發稿日期：114年6月30日

聯絡人：周貝倫科長

聯絡電話：0952-808-437

近日氣溫攀升，各地氣溫炎熱，今（30）日中央氣象署高溫資訊顯示臺北市為橙色燈號，

https://www.gov.taipei/News_Content.aspx?n=F0DDAF49B89F9413&sms=72544237BBF4C5F6&s=9D439F12A1ACF489

Oceans

2025/01/05
澳洲

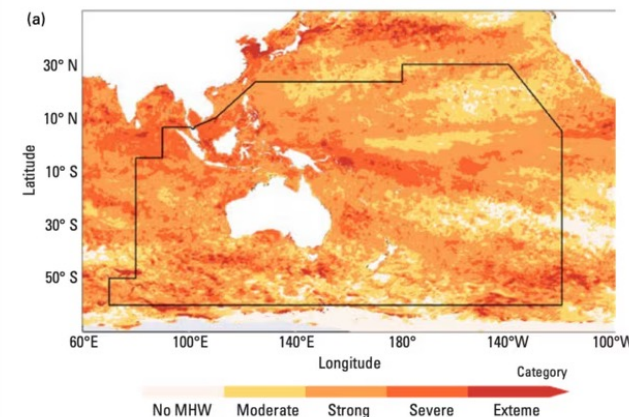
Ibrahim Readfearn

Thu 5 Jun 2025 01:01 BST

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Marine heatwave found to have engulfed area of ocean five times the size of Australia

World Meteorological Organization report says record heat in 2024 was driven by climate crisis and intersected with extreme weather events



<https://www.theguardian.com/environment/2025/jun/05/marine-heatwave-found-to-have-engulfed-area-of-ocean-five-times-the-size-of-australia>

23年夏の北日本の記録的高温 「海洋熱波」が影響か

災害・気象

+ フォローする

2024年7月19日 18:00 [会員限定記事]

2023
日本

保存

✉ n ✕ f ↗

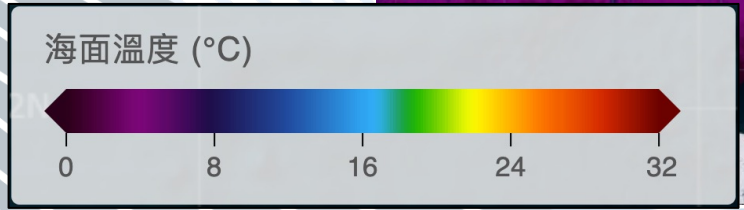
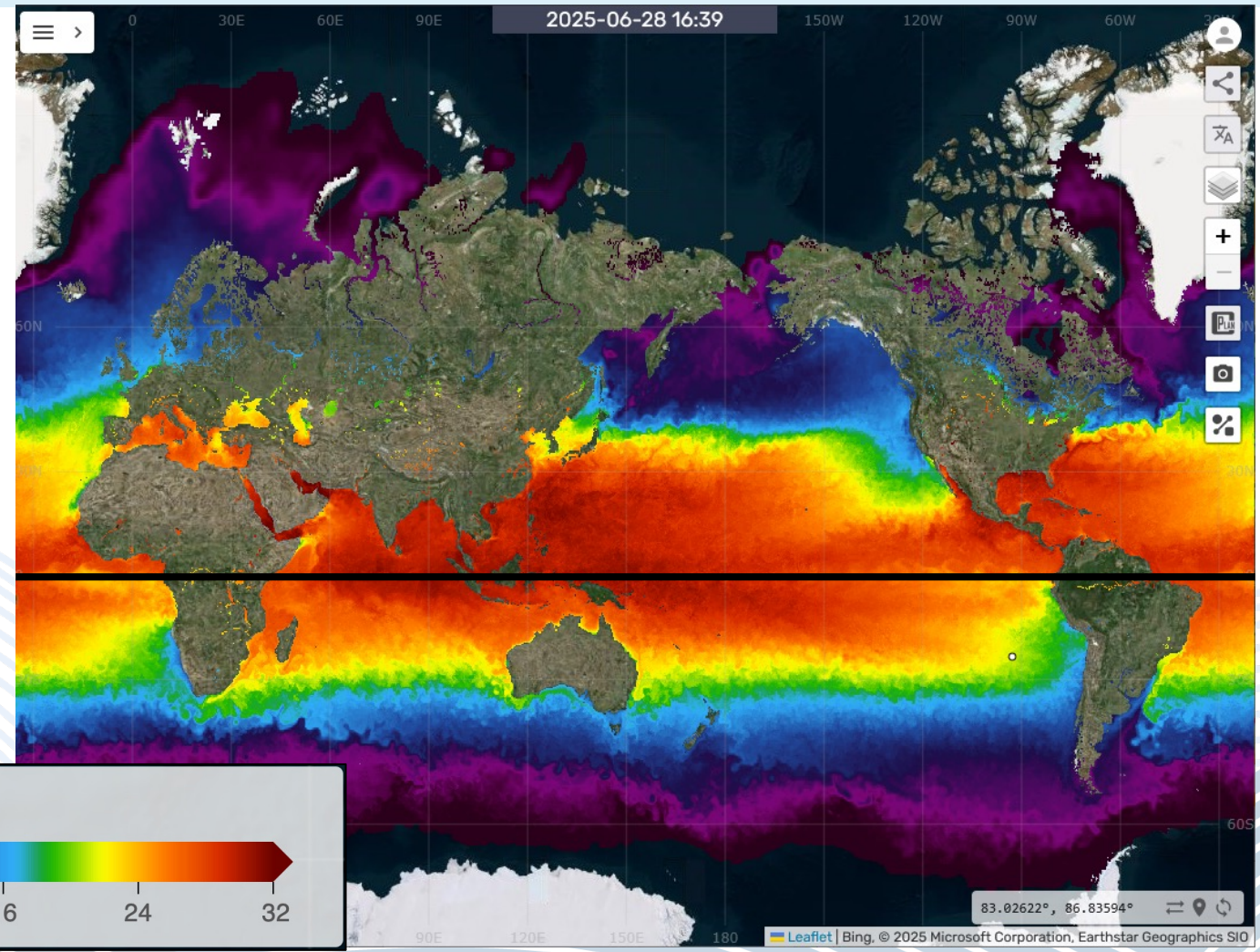
気象庁などは19日、2023年夏の北日本の記録的猛暑について、近海の水温が平年を大きく上回る「海洋熱波」が影響した可能性が高いとの共同研究結果を公表した。

23年夏の日本の平均気温は1898年の統計開始以来最も高く、北日本の太平洋側は平年より3度以上高いエリアもあった。海面水温は暖かい海流の黒潮が北上したことなどから平年より4度以上高く、気温への影響が指摘されていた。

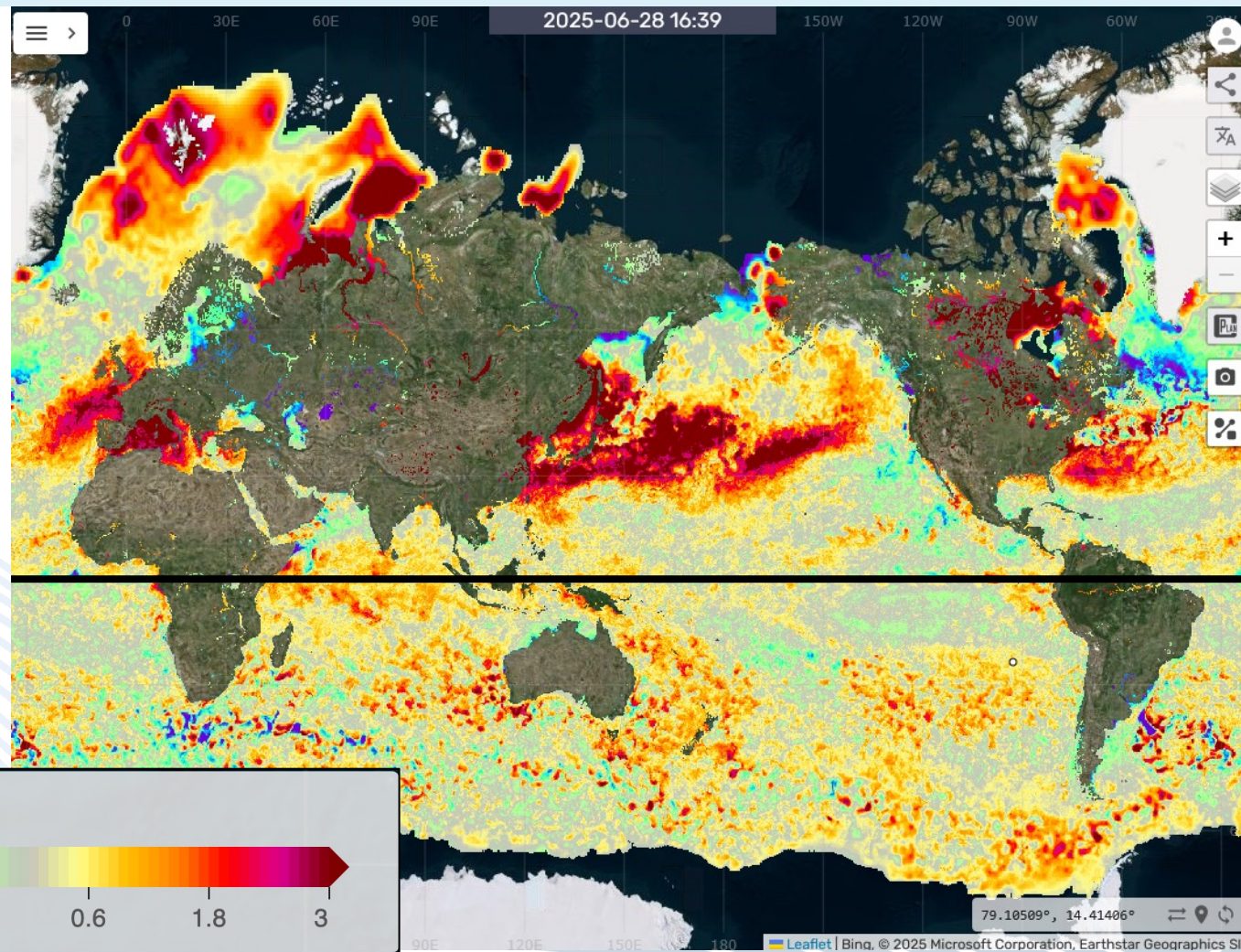
気象庁や専門家らがデータを...

<https://www.nikkei.com/article/DGXZQOUE190VV0Z10C24A7>

Hidy NASA GIBS GHRSSST 海面溫度



Hidy NASA GIBS GHRSSST Anomaly 海面溫度距平



0°

現在氣溫 - 平均氣候值 (climatology, 2003-2014) = 距平值

從文獻到實作

從 Jacox 的研究出發

Article

Thermal displacement by marine heatwaves

<https://doi.org/10.1038/s41586-020-2534-z>

Received: 28 January 2020

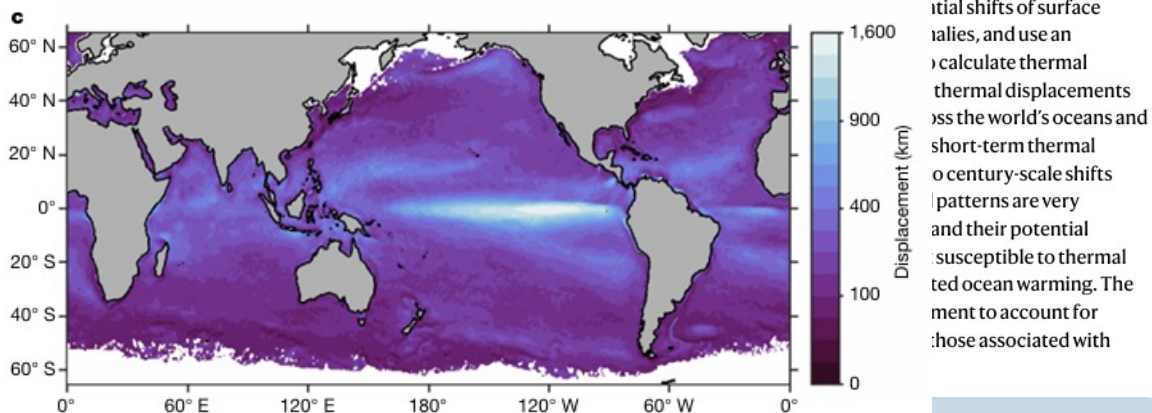
Accepted: 15 June 2020

Published online: 5 August 2020

 Check for updates

Michael G. Jacox^{1,2,3}✉, Michael A. Alexander², Steven J. Bograd^{1,3} & James D. Scott^{2,4}

Marine heatwaves (MHWs)—discrete but prolonged periods of anomalously warm ocean temperatures—can drastically alter ocean ecosystems, with profound ecological and socioeconomic impacts^{1–8}. Considerable effort has been directed at understanding the patterns, drivers and trends of MHWs globally^{9–11}. Typically, MHWs are characterized on the basis of their intensity and persistence at a given location—an approach that is particularly relevant for corals and other sessile organisms that must endure increased temperatures. However, many ecologically and commercially important marine species respond to environmental disruptions by relocating to favourable habitats, and dramatic range shifts of mobile marine species are among the conspicuous impacts of MHWs^{1,4,12,13}. Whereas spatial temperature shifts have been studied extensively in the context of long-term warming trends^{14–18}, they are unaccounted for in existing global MHW analyses. Here we introduce thermal



(Jacox et al., 2020)



國科會海洋學門資料庫 2025/07/17 @ 諾大師-海洋大數據競賽

- 海洋熱浪會讓海洋生物空間的分佈發生改變
- 需要判斷
 - 什麼時候、哪裡出現了熱浪？
 - 若有熱浪
 - 計算生物為了要回到長期平均海表溫度所需移動的最小距離 (thermal displacement)
- 研究結果
- 熱浪
 - 定義 (Jacox et al., 2020)
 - 分級 (Hobday et al., 2018)

熱浪定義的三個元素

- 時間解析度：月
- 定義：每月海表溫度的距平值至少高於季節背景值的第 90 個百分位數。
 - 距平值
 - 背景值
 - 門檻值

熱浪定義：超過門檻的異常高溫

- 時間解析度：月
- 定義：每月海表溫度的距平值至少高於季節背景值的第 90 個百分位數。

- 距平值

- 背景值

- 門檻值

距平值 = 實測溫度 - 氣候值 (長期平均溫度1982~2011)

計算距平值前，還需要計算氣候值

熱浪定義：超過門檻的異常高溫

- 時間解析度：月
- 定義：每月海表溫度的距平值至少高於季節背景值的第 90 個百分位數。

- 距平值

- 背景值—季節背景值

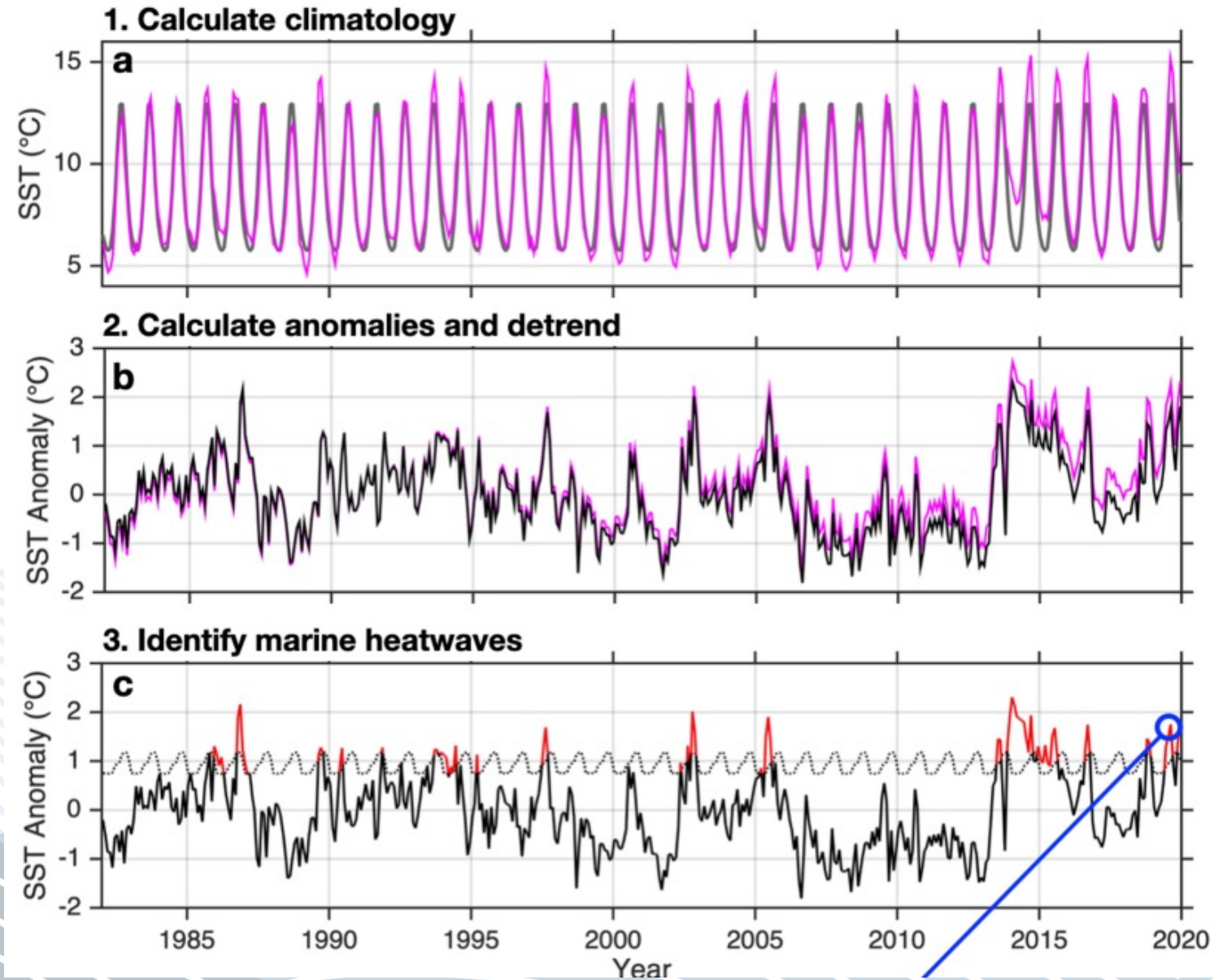
- 門檻值—90%



季節背景值：將計算月份的前後一個月納入當作背景值

Ex. 想判斷 2 月是否有海洋熱浪
--> 綜合過去 1982-2011 年(氣候值)中 1、2、3 月全部每月 SST 距平值的第 90 個百分位數。

(Jacox al., 202



同一個地點
計算氣候背景值

計算距平值

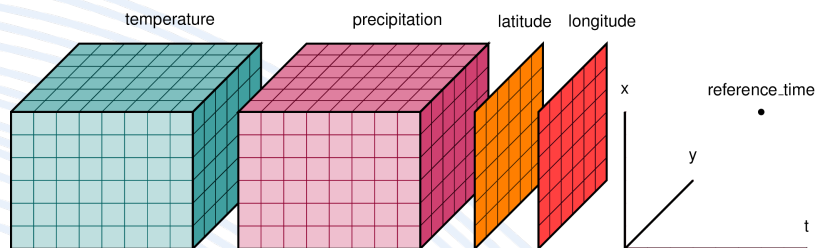
計算門檻 &
判斷是否為海洋熱浪

Coding Time



Xarray for NetCDF

- 海溫的資料是 NetCDF 檔案格式
- 空間解析度： $0.25^{\circ} \times 0.25^{\circ}$
- 處理多維度的矩陣資料



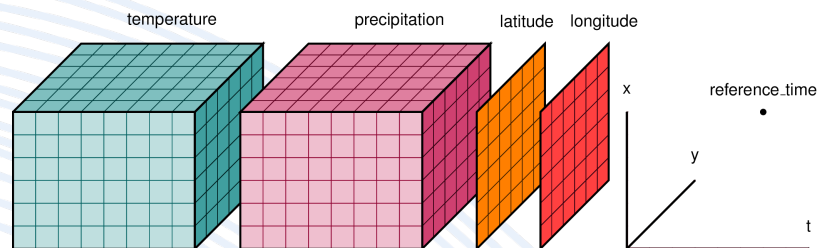
```
import xarray
ds0 =
xarray.open_dataset("./data/oisst-
avhrr-v02r01.20250501.nc")

print(ds0) #印出表頭
```

```
<xarray.Dataset> Size: 17MB
Dimensions: (time: 1, zlev: 1, lat: 720, lon: 1440)
Coordinates:
  * time      (time) datetime64[ns] 8B 2025-05-01T12:00:00
  * zlev      (zlev) float32 4B 0.0
  * lat       (lat) float32 3kB -89.88 -89.62 -89.38 -89.12 ... 89.38 89.62 89.88
  * lon       (lon) float32 6kB 0.125 0.375 0.625 0.875 ... 359.4 359.6 359.9
Data variables:
  sst         (time, zlev, lat, lon) float32 4MB ...
  anom        (time, zlev, lat, lon) float32 4MB ...
  err         (time, zlev, lat, lon) float32 4MB ...
  ice         (time, zlev, lat, lon) float32 4MB ...
Attributes: (12/37)
  Conventions: CF-1.6, ACDD-1.3
  title: NOAA/NCEI 1/4 Degree Daily Optimum Interpolat...
  references: Reynolds, et al.(2007) Daily High-Resolution-...
  source: ICOADS, NCEP_GTS, GSFC_ICE, NCEP_ICE, Pathfin...
  id: oisst-avhrr-v02r01.20250501.nc
  naming_authority: gov.noaa.ncei
  ...
  time_coverage_start: 2025-05-01T00:00:00Z
  time_coverage_end: 2025-05-01T23:59:59Z
  metadata_link: https://doi.org/10.25921/RE9P-PT57
  ncei_template_version: NCEI_NetCDF_Grid_Template_v2.0
  comment: Data was converted from NetCDF-3 to NetCDF-4 ...
  sensor: Thermometer, AVHRR
```

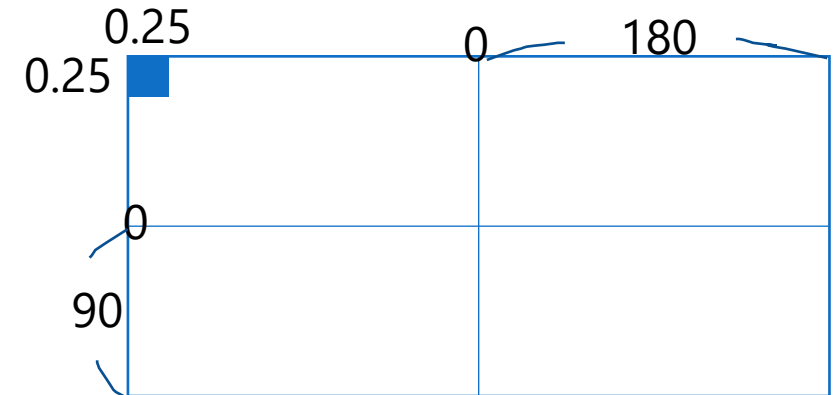
Xarray for NetCDF

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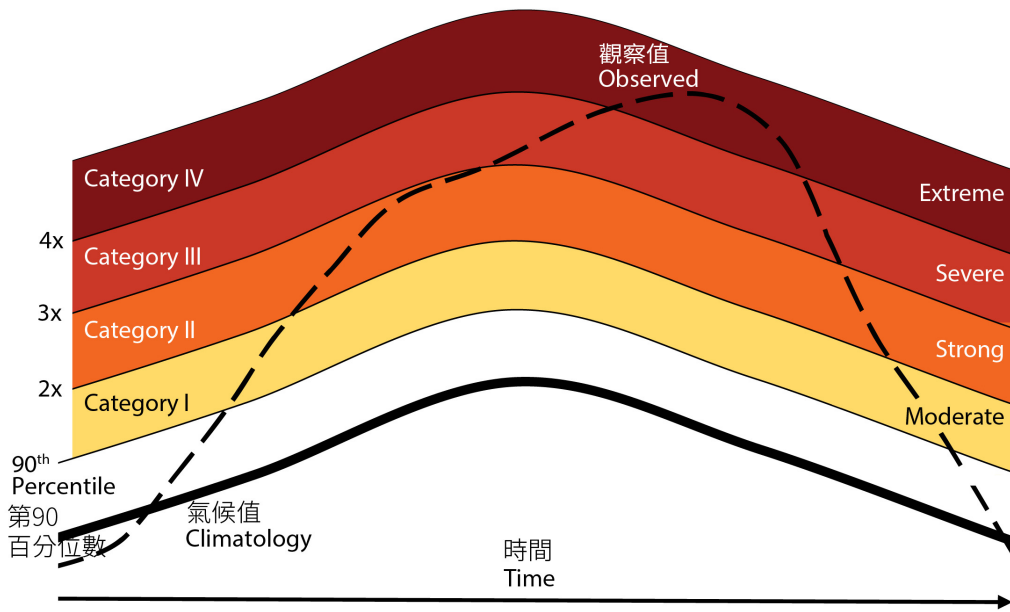


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<xarray.Dataset> Size: 17MB
Dimensions: (time: 1, zlev: 1, lat: 720, lon: 1440)
Coordinates:
  * time      (time) datetime64[ns] 8B 2025-05-01T12:00:00
  * zlev      (zlev) float32 4B 0.0
  * lat       (lat) float32 3kB -89.88 -89.62 -89.38 -89.12 ... 89.38 89.62 89.88
  * lon       (lon) float32 6kB 0.125 0.375 0.625 0.875 ... 359.4 359.6 359.9
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  comment: Data was converted from NetCDF-3 to NetCDF-4 ...
  sensor: Thermometer, AVHRR
```

熱浪分級定義🔥

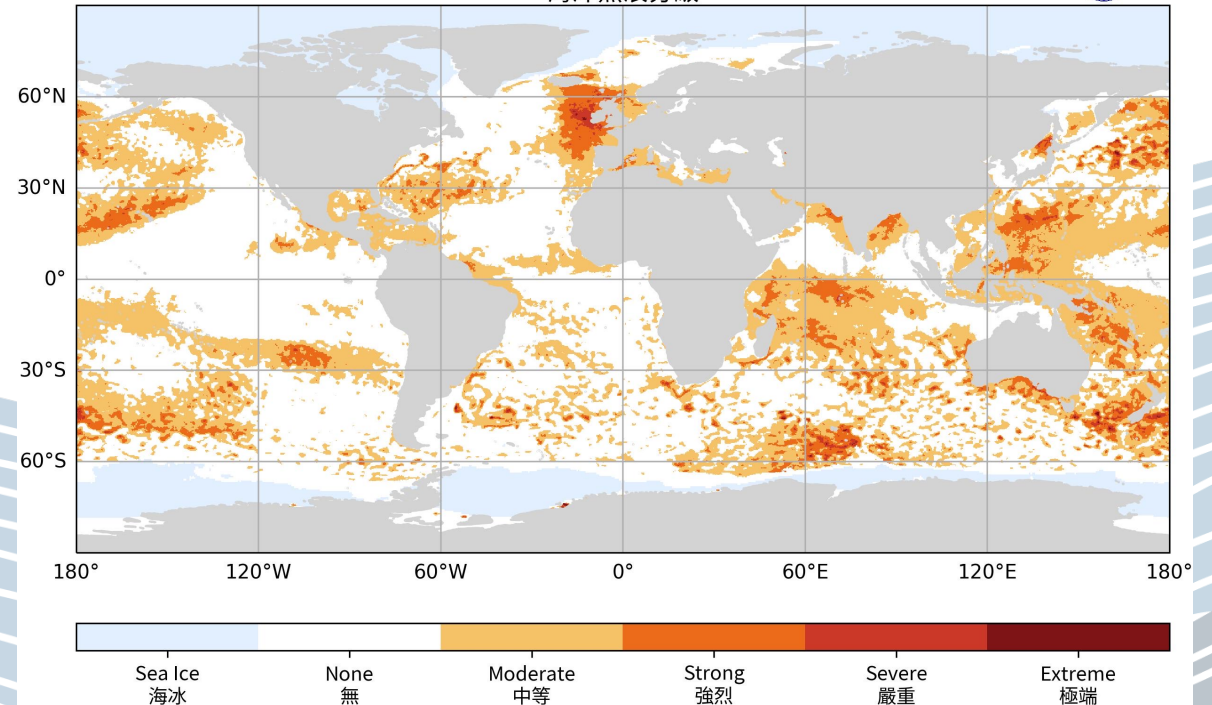
熱浪分級定義

- 中等 (Moderate)
- 強烈 (Strong)
- 嚴重 (Severe)
- 極端 (Extreme)



Hobday et al., 2018

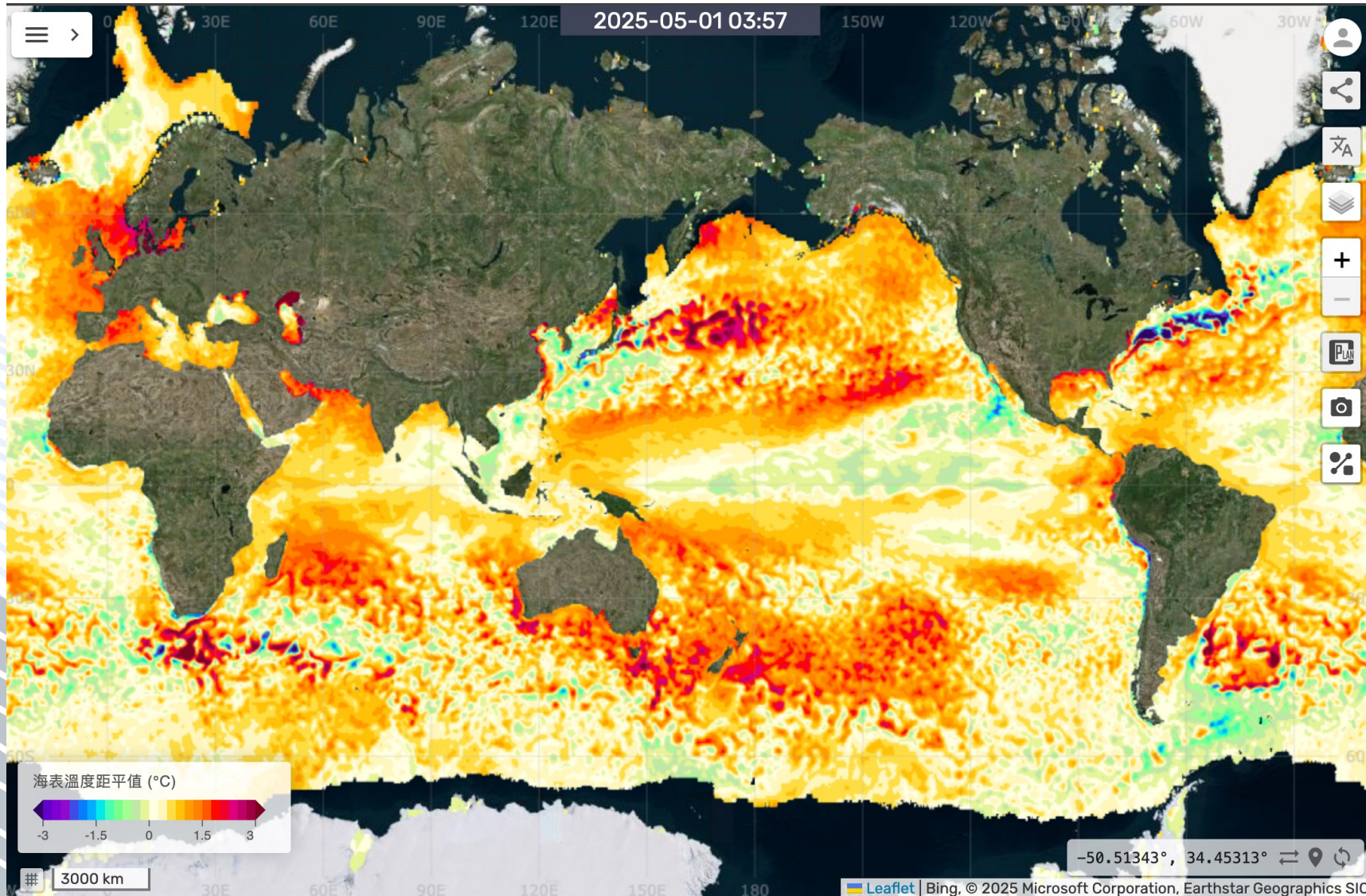
2025/05 Marine Heatwave levels (Monthly 25km SST Anomaly)
海洋熱浪分級

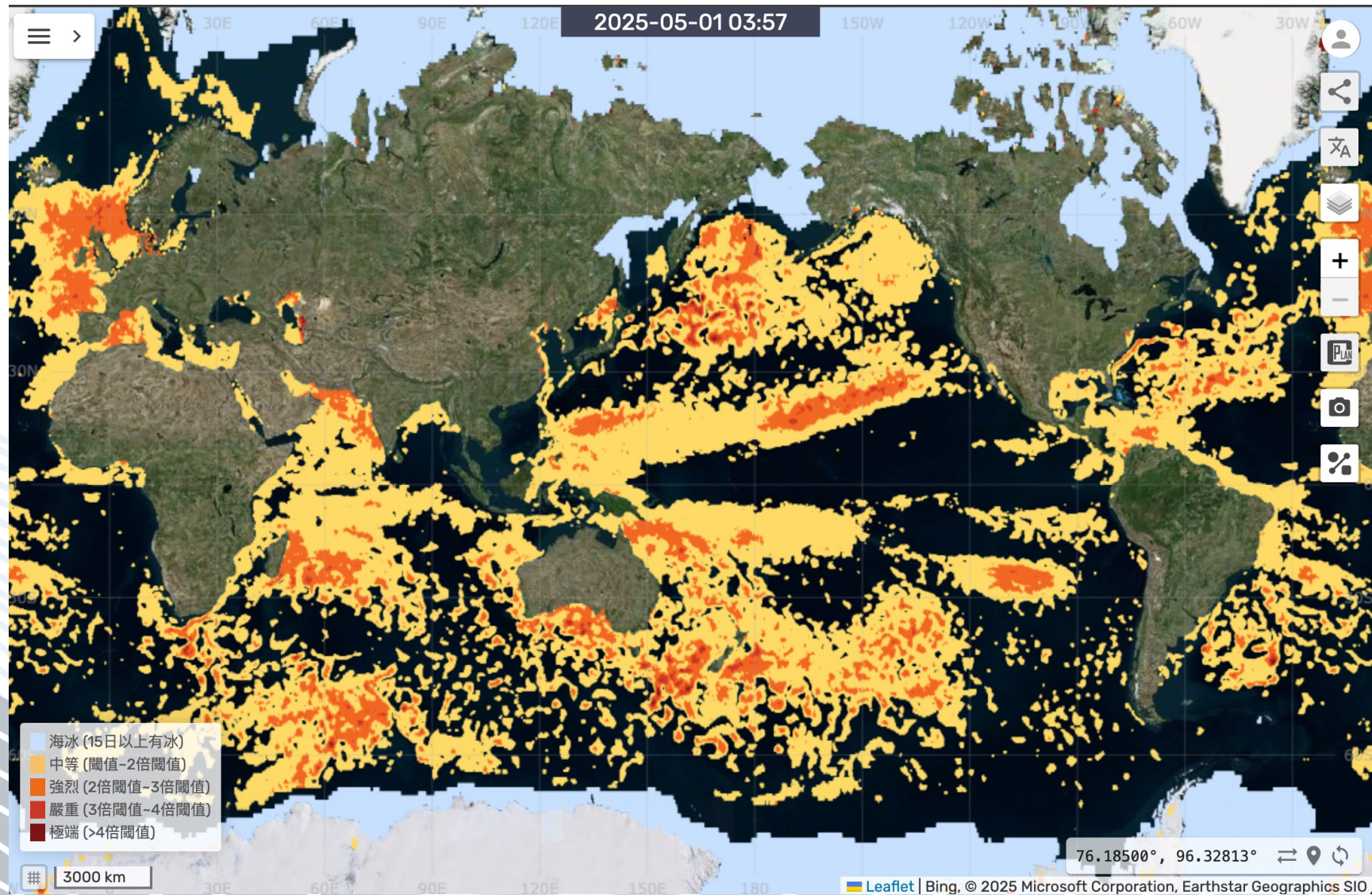


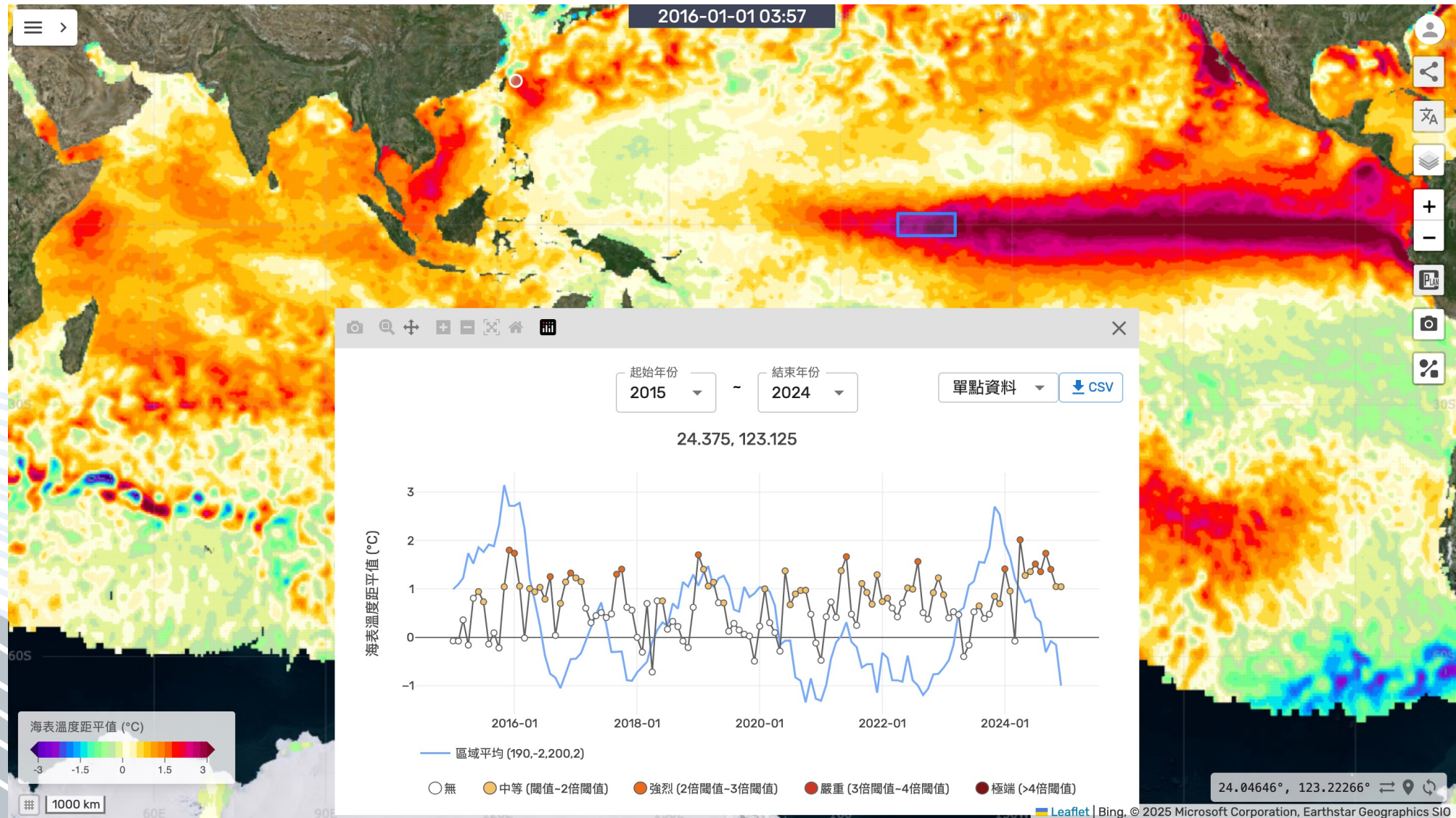
Hidy Viewer



<https://odbview.oc.ntu.edu.tw/hidy/>







參考網頁 👍



- ODB 海洋熱浪

<https://eco.odb.ntu.edu.tw/pub/MHW/>

- Hidy Viewer

<https://odbview.oc.ntu.edu.tw/hidy/>

- NOAA 海溫資料 (OISST)

<https://www.ncei.noaa.gov/products/optimum-interpolation-sst>

- NOAA Coral Reef Watch

<https://coralreefwatch.noaa.gov/>